Zone 5 Technical Overview

Smart i-LAB Zone 5

Dashboard, API context, model contract, CI/CD, deployment, and verification

pctiope/smart-i-lab-testbed

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Objective

Review question

How does the Zone 5 stack move from lab devices and CV labels to a live dashboard, model training, staged rollout, and production verification?

Zone 5 only

Repo-grounded

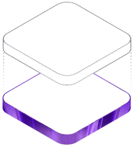
No runtime secrets

- Keep Digital Twin and IoT1 context only where it explains Zone 5 inputs.
- Show user flow, data flow, and service ownership boundaries.
- Separate staging Compose, production systemd, retraining, and model delivery.
- Explain the Zone 5 label contract, feature contract, and mmWave recency path.

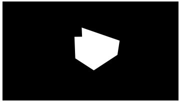
01

Frontend And API Context

Zone 5 Dashboard



Tracked dashboard visual reference.



Desk 5 ROI mask used by the CV counter.

What the operator sees

- Occupancy probability, CV truth, recent history, squared error, and sensor context.
- Annotated MJPEG from the file-backed latest-frame path.
- Split-pane layout with the camera feed beside the readout surface.

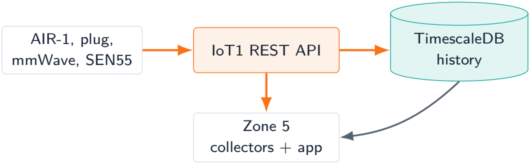
Backend contract

`/api/health`, `/api/current`, `/api/history`, `/api/stream`, and `/api/video.mjpg` are preserved so UI changes do not redefine runtime APIs.

Shared API Context

Purpose

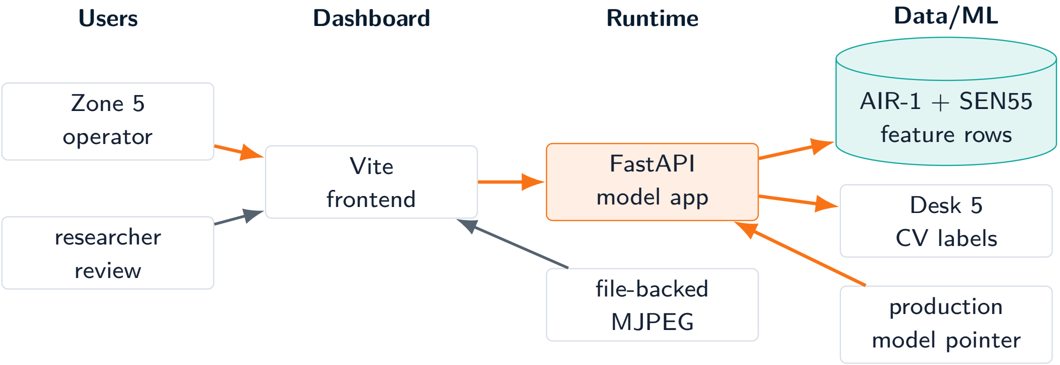
- IoT1 REST API is the integration layer for AIR-1 sensor history and device metadata.
- Zone 5 consumes API data through package-local collectors and live data sources.
- Runtime clients send `x-api-key`; populated values are kept outside source.



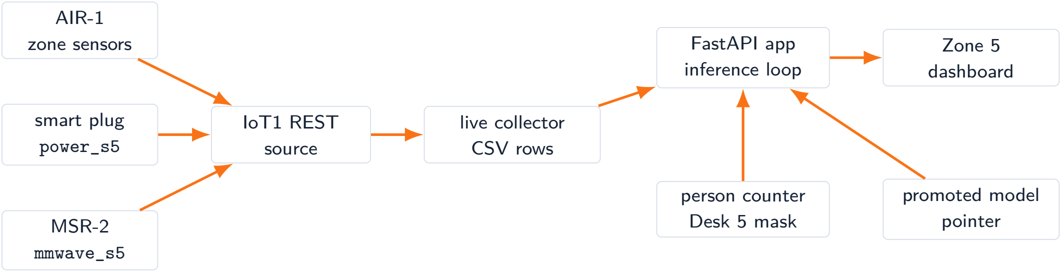
02

Zone 5 Architecture

Zone 5 User And Data Flow



Zone 5 Runtime Architecture

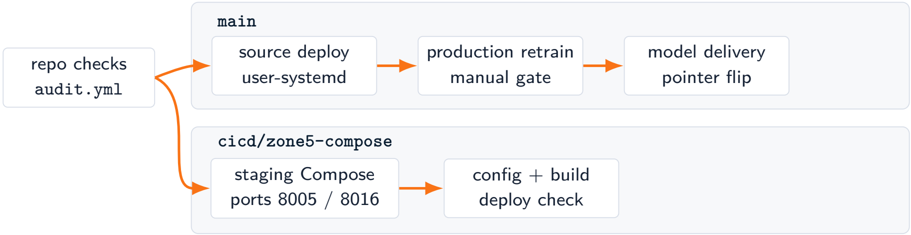


AIR-1 sensors Power mmWave CV labels Model delivery

03

CI/CD And Model Delivery

GitHub Workflow Topology



Workflow	Ownership
`zone5-systemd-source-deploy.yml`	Production source deploy; restarts live services only.
`zone5-compose-ci-cd.yml`	Staging Compose path; self-hosted deploy to separate ports.
`zone5-production-retrain.yml`	Starts the trainer and packages a promoted model only after gating.
`zone5-model-delivery.yml`	Installs a vetted model artifact and updates the production pointer.

Staging Compose Vs. Production systemd

Production

- Branch: `main`
- Runtime: user-level systemd
- Ports: backend 8000, Vite 8015
- Source deploy refuses active trainer collisions.

Staging

- Branch: `cicd/zone5-compose`
- Runtime: Docker Compose
- Ports: backend 8005, Vite 8016
- Copies env and model pointer from production when needed.

Source deploy, staging deploy, training, and model delivery stay separate.

Retrain And Model Delivery

manual / gated trigger → `zone5-trainer` oneshot → stable CSV snapshot → train + validate checks → model artifact → production pointer

- Scheduled retraining is present but disabled until the manual proof run is accepted.
- Promotion failure leaves collection and serving alone.
- Sanitized model registry metadata can be committed; generated run directories stay off-source.

04

Labels And Methodology

Zone 5 Labels

CV truth

- One camera, one Desk 5 mask, one 10-second Zone 5 label stream.
- Target is `zone_occupied`.
- Occupied when median person count crosses the configured threshold.

Guardrails

- Missing labels stay null until a real CV bucket exists.
- Sensor missingness is diagnostic metadata, not an occupancy signal.
- Trainer snapshots the rolling CSV before fitting.

RTSP frames → Desk 5 mask + YOLO → 10-second buckets → `zone_occupied` target

Zone 5 Methodology

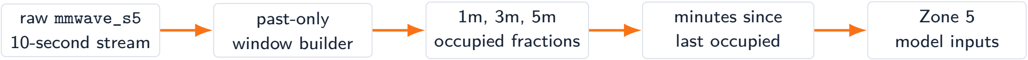
Feature families

- AIR-1 Zone 5: temperature, humidity, CO2, PM2.5.
- Smart plug: `power_s5`.
- mmWave: `mmwave_s5` and recency context.
- SEN55: particulate, environment, VOC, NOx.
- Time: hour and day-of-week cycles.

Training contract

- Current marker: `zone5_mmwave_recency_v1`.
- Missing indicators are diagnostics and metadata, not model inputs.
- Legacy `zone5_missingness_decoupled_v1` artifacts remain loadable for live compatibility.
- Core live gating covers AIR-1, `power_s5`, and `mmwave_s5`; SEN55 is optional.

mmWave Recency Contract



- mmWave remains an input feature, not the training target.
- Recency features are computed from prior samples only.
- The model sees short-history motion context without future leakage.
- Dashboard history surfaces the lookback context beside probability.

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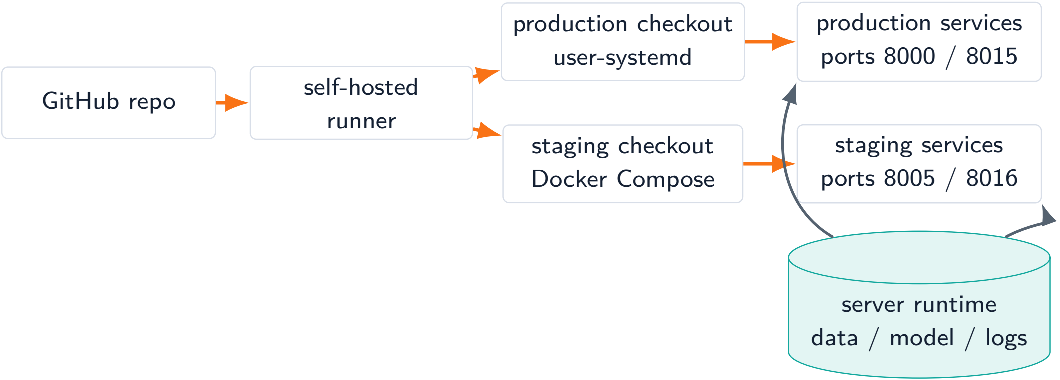
Deployment And Verification

Deployment Architecture



Runtime credentials, generated data, model runs, and logs live on the server, not in the presentation or source history.

Verification And Roadmap

Verification gates

- Python compile checks for package and app modules.
- Zone 5 ground-truth contract tests.
- Vite build checks.
- Systemd unit verification.
- Docker Compose config and image build.
- `/api/health` confirms app health and model readiness fields.

Roadmap and risks

- Prove manual Zone 5 retraining before enabling scheduled retrain.
- Add model freshness, promotion, and upstream API latency monitoring.
- Live QA depends on lab network, API, cameras, and devices.
- Training quality depends on both-class CV labels across usable windows.

Source Anchors

Topic	Primary source paths
Dashboard and runtime	`zone5_cv_time_features_package/web_app/`, `zone5/.../web_app_vite/`
Labels and model contract	`zone5_cv_time_features_package/PIPELINE.md`, `zone5/.../feature_contract.py`
mmWave parsing	`zone5/.../air1_exporter.py`, `zone5/.../feature_builder.py`
CI/CD and deployment	`.github/workflows/zone5-*.yml`, `zone5/.../CICD_SYSTEMD.md`, `zone5/.../CICD_DOCKER_COMPOSE.md`
Model delivery	`zone5/.../CICD_MODEL_DELIVERY.md`, `zone5/.../model_registry/README.md`

Closing point: source, staging runtime, production runtime, training, and model artifacts each have a separate owner and gate.